

02

EVENT TICKETING SYSTEM

Problem Statement: Blockchain-Based Event Ticketing System

Problem Explanation:

Event ticket fraud, scalping, and opaque resale markets cause financial losses and frustration for both organizers and attendees. Centralized ticketing platforms often fail to ensure authenticity and fair pricing.

You are tasked with creating a blockchain-powered ticketing solution (preferably a web application interacting with smart contracts deployed on the test net), where each ticket is minted as a unique NFT. This makes duplication impossible and provides verifiable ownership. Include a secondary marketplace with enforced price controls and royalties on every resale so that event organizers always receive a percentage, preventing exploitative markups and ensuring authenticity.

Resources for Beginners:

- **Basic Malware Analysis: Malware Unicorn's Static Analysis Tools:**
- **YARA: Pattern matching Pefile: Windows PE analysis**
- **Web Development Basics (if unfamiliar): MDN Web Docs W3Schools**

Evaluation Metrics:

A GitHub repository link is expected as your submission, containing a README file with clear instructions on how to run your code. (mandatory).

1. **Integrity & Authenticity:** Each ticket's on-chain metadata must be tamper-proof.
2. **Scalability:** The number of transactions the system can handle (important for large events).
3. **User Experience:** Simplicity of buying, selling, and transferring tickets even for non-crypto-savvy users.
4. **Fair Pricing:** Mechanisms to prevent scalping and excessive markups.
5. **Smart Contract Quality:** Security, clarity, and efficiency of the code.

Examples :

- **Concert Ticket Purchase:** Alice buys a ticket on the platform and holds the NFT in her wallet. On event day, the QR code linked to the NFT is scanned for entry.
- **Resale:** Bob wants to resell his ticket; the on-chain contract enforces a maximum resale price and routes a royalty to the event organizer automatically.



What We Expect:

- **Blockchain Layer:** Ethereum, Polygon, or other EVM-compatible chains.
- **Smart Contract Development:** Solidity (using frameworks like Hardhat or Truffle).
- **Frontend:** React or Vue with Web3 libraries (ethers.js/web3.js).
- **NFT Standards:** ERC-721 or ERC-1155 for ticket tokens.
- **Optional:** Payment gateway integration for fiat on-ramps (e.g., Stripe crypto).

Additional Suggestions:

- **Bonus Features:**
 - **Bidding System:** Auction-based approach for premium seats.
 - **Anti-Bot Measures:** Smart contract-based purchase limits.
 - **Dynamic Ticketing:** Update seat details or perks post-mint without reissuing.

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